Armaan Amatya

(346) 434-0693 | armaanamatya2014@gmail.com | linkedin.com/in/armaan-amatya | armaanamatya.site

Education

University of Houston Houston, TX
Bachelor of Science in Computer Science and Mathematics *Aug. 2024 – Dec. 2026*

Experience

Autonomize AI Austin, TX
Machine Learning Engineer Intern *May 2026 – Present*

- Building deep RL algorithms to improve LLM alignment in clinical healthcare workflows on the AI/ML team
- Automated processing of 30k+ claims weekly by building multimodal document-intelligence pipelines combining OCR, layout parsing, and LLM-based extraction over unstructured clinical records
- Shipped multi-agent LLM systems from POC to production for Fortune 500 healthcare customers with human-in-the-loop evaluation and auditability

DeepInvent.ai (#5 hire) Austin, TX
AI Engineer *May 2026 – Present*

- Enabled 2,000+ patentable inventions in launch week by building the evaluation stack for AI research agents that mine scientific literature and patent filings at a pre-seed autoresearch startup
- Blocked regressions across model, prompt, and tool changes by engineering replay-based eval harnesses with golden-set suites and calibrated LLM-as-judge graders scoring invention novelty

University of Houston Houston, TX
Undergraduate Researcher *Jan. 2026 – Present*

- Cut multimodal LLM inference latency ∼40% by developing a novel token compression method reducing visual tokens ∼60% while retaining ∼98% of baseline benchmark accuracy on Qwen2.5-Omni and Qwen3-Omni
- Deployed compressed models to NVIDIA Jetson Orin for on-device inference after multi-GPU distributed training (PyTorch DDP) on RTX 6000s, cutting VRAM ∼45% and sustaining ∼2x higher tokens/s; won Best Research Award

Artinafti Austin, TX
AI Engineer Intern *Jan. 2026 – May 2026*

- Improved visual fidelity 25% by fine-tuning diffusion-based upscaling and generation models with LoRA adapters and detail-preserving objectives for text and faces, adding photographic color-tone control
- Reduced inference latency 30% by building scalable model-serving APIs with FastAPI, Redis, and PostgreSQL with request batching and caching

AutoHDR Austin, TX (Remote)
Machine Learning Engineer Intern *Jan. 2026 – Apr. 2026*

- Cut per-image turnaround from ∼5 minutes to under 15 seconds across 10,000+ listing photos monthly with diffusion image-to-image pipelines (PyTorch) for exposure fusion, denoising, and 4x super-resolution
- Reduced redundant processing ∼25% by training an image-deduplication model, contrastively fine-tuning CLIP embeddings on listing photo pairs to catch near-duplicate shots
- Produced 1,000+ property tour videos monthly by shipping an image-to-video generation pipeline with depth-aware camera motion and temporally consistent shot transitions

FuseMachines New York, NY
Machine Learning Engineer Intern *May 2024 – Aug. 2024*

- Achieved 98% precision on real-world attendance tracking by fine-tuning a FaceNet model; served real-time inference to a production mobile app (FastAPI, Docker) with async batched requests

Projects

Mixed-Precision GEMM in Triton | *Triton, CUDA, PyTorch, Nsight Compute*

- Built and autotuned a tiled FP16 GEMM kernel in Triton; benchmarked against cuBLAS across RTX/RX GPUs, profiling occupancy and memory bandwidth with Nsight Compute

QLoRA Fine-Tuning + RAG for Scientific QA | *HuggingFace, PEFT, BitsAndBytes, FAISS, LangChain*

- Fine-tuned Mistral-7B with QLoRA (4-bit NF4) on 234 curated QA pairs; built hybrid RAG over a 75-paper FAISS corpus; evaluated with GPT-4o LLM-as-judge; shipped quantized inference on HuggingFace Spaces

Technical Skills

Languages: Python, C++, CUDA, Rust, TypeScript, SQL, Bash
ML Systems: PyTorch, Triton, vLLM, TensorRT, ONNX, HuggingFace, quantization (4-bit/QLoRA), model compression, FAISS, Nsight
AI/ML: LLMs, RL (PPO/GRPO/RLHF), multimodal models, diffusion models, RAG, distributed inference, Docker, Kubernetes